

AI & Machine Learning Internship

Task 2: Feature Scaling and Model Comparison Using California Housing Dataset

Submitted By

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1. Objective

The objective of this task is to compare the performance of multiple regression models using the California Housing Dataset. The models were trained to predict house prices based on housing features. Feature scaling was applied where required, and the models were evaluated using Root Mean Squared Error (RMSE) and R^2 Score to determine the best-performing model.

2. Dataset Description

The California Housing Dataset is a real-world dataset available in the Scikit-Learn library. It contains information collected from California housing districts.

The dataset includes the following features:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable is the median house value.

3. Methodology

The following steps were performed during the implementation:

- Imported the required Python libraries.
- Loaded the California Housing Dataset.
- Converted the dataset into a Pandas Data Frame.

- Split the dataset into training and testing sets using an 80:20 ratio.
- Applied StandardScaler for feature scaling on Linear Regression and Ridge Regression.
- Trained three regression models:
 - Linear Regression
 - Ridge Regression
 - Decision Tree Regressor
- Evaluated each model using RMSE and R^2 Score.
- Compared the models using a performance table and bar chart.
- Visualized the predictions using an Actual vs Predicted scatter plot.

4. Results

The models were compared based on prediction accuracy.

Model	Observation
Linear Regression	Good baseline performance
Ridge Regression	Similar performance to Linear Regression with regularization
Decision Tree Regressor	Best overall performance with the highest R^2 Score and lowest RMSE

Performance Summary

- Decision Tree RMSE: **0.703**
- Decision Tree R^2 Score: **0.623**

The Decision Tree Regressor produced more accurate predictions than the other two models.



5. Conclusion

In this task, three regression models were successfully trained and evaluated using the California Housing Dataset.

Feature scaling was applied using StandardScaler for Linear Regression and Ridge Regression models. The models were evaluated using RMSE and R^2 Score.

Among all the models, the **Decision Tree Regressor** achieved the best performance by obtaining the **lowest RMSE** and the **highest R^2 Score**. This indicates that it was able to predict house prices more accurately than the other models.

This task provided practical experience in data preprocessing, feature scaling, regression modelling, model evaluation, and performance comparison. Future improvements may include the use of ensemble learning methods such as Random Forest, Gradient Boosting, and hyperparameter tuning to further improve prediction accuracy.

6. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

7. Outcome

The objectives of Task 2 were successfully achieved. Multiple machine learning regression models were implemented, evaluated, and compared. The Decision Tree Regressor was identified as the most suitable model for this dataset based on its superior prediction performance.